q1- hacker earth

All Tracks > Basic Programming > Input/Output > Basics of Input/Output > Problem

Read from STDIN, Write to STDOUT

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Problem

Read different types of data from standard input, process them as shown in output format and print the answer to standard output.

Input format:
First line contains integer N.
Second line contains string S.

Output format:
First line should contain N x 2.
Second line should contain the same string S.

Constraints:
$0 \leq N \leq 10$
$1 \leq |S| \leq 15$ where S length of string S

Sample Input	Sample Output
5 helloworld	10 helloworld

Time Limit: 1
Memory Limit: 256

Enter your code or Upload your code as file. Save C++17 (g++ 10.3.0)

```cpp
#include <iostream>
#include<string>
using namespace std;
int main() {
    int num;
    cin >> num;    //Reading input from STDIN
    string a;
    cin>>a;
     cout<<num*2<<endl;
     cout<<a<<endl;
     return 0;

}
```

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Time Limit: 1
Memory Limit: 256
Source Limit:

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Submission ID: 127057360

RESULT: Accepted Refer judge environment

Score	Time (sec)	Memory (KiB)	Language
0	0.00901	2	C++17

q2- ZOOS problem

```cpp
#include <stdio.h>
#include <string.h>

int main() {
char word[100000];
scanf("%s", word);
```

```cpp
int znum = 0, onum = 0;
```

```cpp
for (int i = 0; word[i] != '\0'; i++) {
```

```
if (word[i] == 'z')
znum++;
else if (word[i] == 'o')
onum++;
}
```

```
if (onum == 2 * znum)
printf("Yes");
else
printf("No");
```

```
return 0;
}
```

Problem

You are required to enter a word that consists of x and y that denote the number of **Z**s and **O**s respectively. The input word is considered similar to word **zoo** if $2 \times x = y$.

Determine if the entered word is similar to word **zoo**.

For example, words such as **zzoooo** and **zzzoooooo** are similar to word **zoo** but not the words such as **zzooo** and **zzzoooooo**.

Input format

- First line: A word that starts with several **Z**s and continues by several **O**s. **Note**: The maximum length of this word must be 20.

Output format

Print **Yes** if the input word can be considered as the string **zoo** otherwise, print **No**.

Sample Input	Sample Output
zzzoooooo	Yes

Time Limit: 0.5
Memory Limit: 256
Source Limit:

Explanation

-

Contributers:

Amirreza Poorakhavan

24:1 vscode

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Submission ID: 127057412

RESULT: Accepted | Refer judge environment

Score	Time (sec)	Memory (KiB)	Language
0	0.05626	2	C

Input	Result	Time (sec)	Memory (KiB)	Score	Your output	Correct output	Diff
Input #1	Accepted	0.009259	2	17			
Input #2	Accepted	0.00892	2	17			
Input #3	Accepted	0.009302	2	17			
Input #4	Accepted	0.009477	2	16			
Input #5	Accepted	0.009478	2	16			
Input #6	Accepted	0.009824	2	17			

q3- favourite singer

```
#include <stdio.h>
#include <stdlib.h>

typedef long long ll;
```

```
// Comparator for qsort
```

```
int cmp(const void *a, const void *b) {
ll x = *(ll*)a;
ll y = *(ll*)b;
if (x > y) return 1;
if (x < y) return -1;
return 0;
}
```

```
int main() {
ll n;
scanf("%lld", &n);
```

```
ll *arr = (ll*)malloc(n * sizeof(ll));
```

```
for (ll i = 0; i < n; i++)
scanf("%lld", &arr[i]);
```

```
// Sort the array so identical singers group together
qsort(arr, n, sizeof(ll), cmp);
```

```
ll maxFreq = 1;
ll currFreq = 1;
ll favCount = 0;
```

```
for (ll i = 1; i < n; i++) {
if (arr[i] == arr[i - 1]) {
currFreq++;
} else {
if (currFreq > maxFreq) {
maxFreq = currFreq;
favCount = 1;
} else if (currFreq == maxFreq) {
favCount++;
```

```
}
currFreq = 1;
}
}
```

```
// Check the last block
if (currFreq > maxFreq) {
maxFreq = currFreq;
favCount = 1;
} else if (currFreq == maxFreq) {
favCount++;
}
```

```
printf("%lld\n", favCount);
```

```
free(arr);
return 0;
}
```

Problem

Bob has a playlist of N songs, each song has a singer associated with it (denoted by an integer)

Favourite singer of Bob is the one whose songs are the most on the playlist

Count the number of Favourite Singers of Bob

Input Format

The first line contains an integer N, denoting the number of songs in Bob's playlist.

The following input contains N integers, i^{th} integer denoting the singer of the ith song.

Output Format

Output a single integer, the number of favourite singers of Bob

Note: Use 64 bit data type

Constraints

$1 \leq N \leq 2 * 10^5$
$1 \leq a[i] \leq 10^{15}$

Sample Input	Sample Output
5 1 1 2 2 4	2

Time Limit: 1
Memory Limit: 256
Source Limit:

Score	Time (sec)	Memory (KiB)	Language
0	0.29327	3436	C

Input	Result	Time (sec)	Memory (KiB)	Score	Your output	Correct output	Diff
Input #1	Accepted	0.009961	2	5			
Input #2	Accepted	0.009264	2	5			
Input #3	Accepted	0.009514	2	5			
Input #4	Accepted	0.009128	2	5			
Input #5	Accepted	0.009367	2	5			
Input #6	Accepted	0.009199	2	5			
Input #7	Accepted	0.008766	2	5			
Input #8	Accepted	0.010555	2	10			
Input #9	Accepted	0.058388	3436	10			
Input #10	Accepted	0.049659	3436	10			
Input #11	Accepted	0.0499	3436	20			
Input #12	Accepted	0.009911	2	5			
Input #13	Accepted	0.049656	3436	10			

q4- toggle string

```
#include <stdio.h>
```

```
int main() {
char s[100000];
scanf("%s", s);
```

```
for (int i = 0; s[i] != '\0'; i++) {
```

```
if (s[i] >= 'a' && s[i] <= 'z') {
s[i] = s[i] - 'a' + 'A';
}
```

```
else if (s[i] >= 'A' && s[i] <= 'Z') {
s[i] = s[i] - 'A' + 'a';
}
}
```

```
printf("%s", s);
```

```
return 0;
}
```

Problem

You have been given a String S consisting of uppercase and lowercase English alphabets. You need to change the case of each alphabet in this String. That is, all the uppercase letters should be converted to lowercase and all the lowercase letters should be converted to uppercase. You need to then print the resultant String to output.

Input Format
The first and only line of input contains the String S

Output Format
Print the resultant String on a single line.

Constraints
$1 \leq |S| \leq 100$ *where S denotes the length of string S.*

Sample Input	Sample Output
abcdE	ABCDe

Time Limit: 5
Memory Limit: 256
Source Limit:

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Submission ID: 127057475

RESULT: Accepted — Refer judge environment

Score	Time (sec)	Memory (KiB)	Language
0	0.03689	2	C

Input	Result	Time (sec)	Memory (KiB)	Score	Your output	Correct output	Diff
Input #1	Accepted	0.009427	2	25			
Input #2	Accepted	0.009283	2	25			
Input #3	Accepted	0.008984	2	25			
Input #4	Accepted	0.009194	2	25			

q5- palindromic string

```
#include <iostream>
```

```
#include<string>
using namespace std;
int main() {
string s;
cin>>s;
for(int i=0;i<s.length();i++){
if(s[i]!=s[s.length()-i-1])
{

cout<<"NO";
return 0;
}
}
cout<<"YES";
return 0;
}
```

Problem

You have been given a String *S*. You need to find and print whether this string is a palindrome or not. If yes, print "YES" (without quotes), else print "NO" (without quotes).

Input Format
The first and only line of input contains the String *S*. The String shall consist of lowercase English alphabets only.

Output Format
Print the required answer on a single line.

Constraints $1 \leq |S| \leq 100$

Note
String *S* consists of lowercase English Alphabets only.

Sample Input	Sample Output
aba	YES

Time Limit: 1
Memory Limit: 256
Source Limit:

10:1 vsco

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Submission ID: 127057522

RESULT: Accepted — Refer judge environment

Score	Time (sec)	Memory (KiB)	Language
0	0.03726	2	C++17

Input	Result	Time (sec)	Memory (KiB)	Score	Your output	Correct output	Diff
Input #1	Accepted	0.009784	2	25			
Input #2	Accepted	0.00945	2	25			
Input #3	Accepted	0.009012	2	25			
Input #4	Accepted	0.009014	2	25			

q7- count divisors

```
#include <stdio.h>

int main(){
```

```c
int l,r,k;
scanf("%d %d %d",&l,&r,&k);
int count=0;
for (int i = l; i <= r; i++) {
if (i % k == 0) {
count++;
}
}
printf("%d",count);
return 0;
}
```

oblem

I have been given 3 integers - l, r and k. Find how many numbers between l and r th inclusive) are divisible by k. You do not need to print these numbers, you just e to find their count.

ut Format

first and only line of input contains 3 space separated integers *l*, *r* and *k*.

tput Format

nt the required answer on a single line.

nstraints

$\leq l \leq r \leq 1000$
$\leq k \leq 1000$

ample Input	Sample Output
10 1	10

e Limit: 1
nory Limit: 256
rce Limit:

[redacted]

15:1 vscode

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Submission ID: 127057554

RESULT: Accepted — Refer judge environment

Score	Time (sec)	Memory (KiB)	Language
0	0.03795	2	C

Input	Result	Time (sec)	Memory (KiB)	Score	Your output	Correct output	Diff
Input #1	Accepted	0.00938	2	25			
Input #2	Accepted	0.008964	2	25			
Input #3	Accepted	0.010712	2	25			
Input #4	Accepted	0.00889	2	25			